

Exponentially many f -avoiding flows for groups of order 6 and 7:

Conjecture 1.10 of DeVos, Langhede, Mohar and Šámal

Machine-generated note (see provenance box)

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Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The proof was found by the language model gpt-5.6-sol in a single autonomous pass on 2026-09-01, as part of a large sweep over open problems catalogued from arXiv (catalog id 2005.09767__00; original writeup in attacks/2005.09767__00/output.md). (2) An independent adversarial referee agent (claude-fable-5, 2026-09-02) re-derived every step, checked Theorem 1.9, Conjecture 1.10 and the Jaeger–Linial–Payan–Tarsi input against the source, verified the support bound exhaustively for small parameters, enumerated complete flow spaces on ten 3-edge-connected graphs (with loops and parallel edges) minimising over all f where feasible, and implemented the three $\mathbb{Z}_6$ families literally; every check passed. Its verdict was MINOR_GAPS: one repairable gap (the uniform constant for $|\Gamma| \geq 8$ on graphs with at most 3 vertices, Section 5), a missing attribution (Lemma 2.1), and a degenerate-convention remark. Its report is reproduced verbatim in Appendix B. (3) On 2026-09-09 the writeup was rewritten into the present note by claude-fable-5.1. Three repairs were made, each declared where it occurs: (i) Section 5 derives an explicit uniform constant for $|\Gamma| \geq 8$ from the source’s Theorem 1.9 for $n \geq 4$ and handles $n \leq 3$ by a two-line parallel-edge argument (the writeup asserted the constant without this case analysis); (ii) the support bound is attributed to Alon and Füredi and the Eulerian lift is labelled folklore; (iii) the one-vertex convention is stated and discussed (Remark 7.1). The main constants $7^{1/4}$ and $3^{1/12}$ and all proofs of the $\mathbb{Z}_7$ and $\mathbb{Z}_6$ cases are as in the writeup; full proofs are in Appendix A. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

An oriented graph $G = (V, E)$ is Γ -connected if for every $f: E \rightarrow \Gamma$ there is a Γ -flow ϕ with $\phi(e) \neq f(e)$ for all e . Jaeger, Linial, Payan and Tarsi proved that every 3-edge-connected oriented graph is Γ -connected whenever $|\Gamma| \geq 6$. DeVos, Langhede, Mohar and Šámal showed that for $|\Gamma| \geq 8$ there are exponentially many such flows and conjectured (Conjecture 1.10) that this holds for all $|\Gamma| \geq 6$, the cases $\mathbb{Z}_6$ and $\mathbb{Z}_7$ being open. We prove the conjecture. For $\mathbb{Z}_7$ the number of f -avoiding flows is at least $7^{n/4+1}$, by applying the Alon–Füredi bound on the support of a polynomial over a finite grid to the product of the m linear forms $\phi(e) - f(e)$ on the r -dimensional flow space. For $\mathbb{Z}_6 \cong \mathbb{F}_2 \times \mathbb{F}_3$ the number is at least $3^{n/12+2/3}$: after normalising by one avoiding flow, the forbidden values split into three classes, and three explicit families of admissible flows (binary flows vanishing on one class, ternary flows avoiding one class, and unit-valued lifts of binary flows vanishing on the third) cannot all be small. Together with the source paper’s theorem for $|\Gamma| \geq 8$, this gives the conjecture with the explicit constant $c = 2^{-1/4}(3/2)^{1/2} \approx 1.03$.

1 Introduction

Throughout, following the source paper [3], graphs may have loops and parallel edges. An *oriented graph* $G = (V, E)$ is a graph with a fixed orientation of each edge; we write $n = |V|$, $m = |E|$, and, when G is connected, $r = m - n + 1$ for its cycle rank. For an abelian group Γ , a Γ -flow is a

map $\phi: E \rightarrow \Gamma$ satisfying Kirchhoff's law at every vertex (the sum of the values on outgoing edges equals the sum on incoming edges; a loop contributes equally to both sides and is unconstrained). For $f: E \rightarrow \Gamma$ let

$$N_\Gamma(G, f) = |\{\phi: E \rightarrow \Gamma : \phi \text{ is a flow and } \phi(e) \neq f(e) \text{ for every } e \in E\}|$$

be the number of f -avoiding flows. Following the convention of the writeup, a 3-edge-connected graph is understood to have at least two vertices (see [Remark 7.1](#)); such a graph is connected and has minimum degree at least 3, so

$$m \geq \frac{3}{2}n. \quad (1)$$

Definition 1.1 ([3, Definition 1.8]). Let $G = (V, E)$ be an oriented graph and Γ an abelian group. G is Γ -connected if for every $f: E \rightarrow \Gamma$ there is a flow $\phi: E \rightarrow \Gamma$ with $\phi(e) \neq f(e)$ for every $e \in E$, i.e. if $N_\Gamma(G, f) \geq 1$ for all f .

Theorem 1.2 (Jaeger, Linial, Payan, Tarsi [4]; quoted in [3]). *Every oriented 3-edge-connected graph is Γ -connected whenever $|\Gamma| \geq 6$.*

DeVos, Langhede, Mohar and Šámal count such flows. Their Theorem 1.9 reads verbatim:

Theorem 1.3 ([3, Theorem 1.9]). *Let $G = (V, E)$ be an oriented 3-edge-connected graph with $\ell = |E| - |V|$ and let Γ be an abelian group with $|\Gamma| = k \geq 6$. For every $f: E \rightarrow \Gamma$ we have*

$$N_\Gamma(G, f) \geq \begin{cases} \frac{1}{2} \left(\frac{k-6}{2} \right)^\ell & \text{if } k \text{ is odd,} \\ \frac{1}{2} \left(\frac{k-4}{2} \right)^\ell & \text{if } k \text{ is even.} \end{cases}$$

For $k \geq 8$ this is exponential in ℓ , but for $k \in \{6, 7\}$ it only yields existence. The authors write that they believe this to be a shortcoming of their techniques, and conjecture:

Conjecture 1.4 ([3, Conjecture 1.10]). *There exists a fixed constant $c > 1$ so that the following holds. For every 3-edge-connected oriented graph $G = (V, E)$ of order n , every abelian group Γ with $|\Gamma| \geq 6$, and every $f: E \rightarrow \Gamma$, there exist at least c^n flows $\phi: E \rightarrow \Gamma$ with $\phi(e) \neq f(e)$ for every $e \in E$.*

They note that [Theorem 1.3](#) settles the conjecture for all abelian groups except $\mathbb{Z}_6$ and $\mathbb{Z}_7$, that for $\mathbb{Z}_7$ they have no interesting lower bound, and they prove a superpolynomial bound $2^{\sqrt{\ell}/\log \ell}$ for $\mathbb{Z}_6$ when $\ell \geq 11$ (their Theorem 1.11). Since every abelian group of order 6 is $\mathbb{Z}_6$ and every group of order 7 is $\mathbb{Z}_7$, the two missing cases are exactly these groups.

Theorem 1.5 (Main theorem). *Let $G = (V, E)$ be a 3-edge-connected oriented graph with $n \geq 2$ vertices and m edges, and let $f: E \rightarrow \Gamma$.*

- (i) *If $\Gamma = \mathbb{Z}_7$ then $N_{\mathbb{Z}_7}(G, f) \geq 7^{5m/6-n+1} \geq 7^{n/4+1}$.*
- (ii) *If $\Gamma = \mathbb{Z}_6$ then $N_{\mathbb{Z}_6}(G, f) \geq 3^{(3m-4n+4)/6} \geq 3^{n/12+2/3}$.*
- (iii) *If $|\Gamma| \geq 8$ then $N_\Gamma(G, f) \geq \frac{1}{2}(3/2)^{n/2}$ when $n \geq 4$, and $N_\Gamma(G, f) \geq |\Gamma| - 2 \geq 6$ when $n \in \{2, 3\}$.*

Consequently $N_\Gamma(G, f) \geq c^n$ for every abelian group Γ with $|\Gamma| \geq 6$, with the absolute constant

$$c = \min\{7^{1/4}, 3^{1/12}, 2^{-1/4}(3/2)^{1/2}\} = 2^{-1/4}(3/2)^{1/2} \approx 1.0299.$$

Corollary 1.6. *Conjecture 1.4 holds.*

Proof. Immediate from [Theorem 1.5](#), since every abelian group of order 6 or 7 is cyclic and the case $|\Gamma| \geq 8$ is part (iii). $\square$

Idea of proof. (i) The $\mathbb{F}_7$ -flows form an r -dimensional vector space, $r = m - n + 1$. The f -avoiding flows are exactly the non-zeros of the polynomial $P = \prod_{e \in E} (L_e - f(e))$ of degree m in r variables, where L_e is the coordinate function “value on e ”. P is not identically zero as a function by [Theorem 1.2](#), and the Alon–Füredi bound ([Lemma 2.1](#)) says a non-zero function of degree d on $\mathbb{F}_q^r$ has at least $q^{r-d/(q-1)}$ non-zeros. With $q = 7$, $d = m$ and (1) this gives $7^{5m/6-n+1} \geq 7^{n/4+1}$. (ii) Fix one avoiding flow ϕ_0 ([Theorem 1.2](#)) and translate: the problem becomes counting flows ψ with $\psi(e) \neq t(e) := f(e) - \phi_0(e)$, where t is nowhere zero. Write $\mathbb{Z}_6 \cong \mathbb{F}_2 \times \mathbb{F}_3$ and sort the edges by the value of t into three classes A ($t = 3$), B ($t \in \{2, 4\}$), C ($t \in \{1, 5\}$) of sizes $a + b + c = m$. Binary flows vanishing on A , ternary flows avoiding t on B (counted by [Lemma 2.1](#) over $\mathbb{F}_3$), and unit-valued lifts of binary flows vanishing on C ([Lemma 2.2](#)) are three families of admissible flows of sizes at least $2^{(r-a)_+}$, $3^{(2r-b)_+/2}$ and $2^{(r-c)_+}$. Their exponents sum to at least $4r - m$, so one of them is at least $3^{(4r-m)/6} = 3^{(3m-4n+4)/6} \geq 3^{n/12+2/3}$. (iii) For $|\Gamma| \geq 8$ the base in [Theorem 1.3](#) is at least $3/2$ and $\ell = m - n \geq n/2$, so $N \geq \frac{1}{2}(3/2)^{n/2}$, which exceeds 1 exactly when $n \geq 4$ and then dominates c^n . A 3-edge-connected graph with $n \leq 3$ has two parallel edges; adding $t \in \Gamma$ around that 2-cycle to one avoiding flow produces $|\Gamma|$ distinct flows of which at most two are not avoiding. [\[full proof\]](#)

2 Two tools

Lemma 2.1 (Alon–Füredi support bound). *Let q be a prime power and let $P \in \mathbb{F}_q[X_1, \dots, X_r]$ induce a non-zero function on $\mathbb{F}_q^r$, with $\deg P \leq d$. Then*

$$|\{x \in \mathbb{F}_q^r : P(x) \neq 0\}| \geq q^{(r-d/(q-1))_+}, \quad \text{where } (z)_+ = \max\{z, 0\}.$$

Idea of proof. Attribution (added in the rewrite). This is the bound of Alon and Füredi [1, Theorem 5] on the number of non-zeros of a polynomial on a grid, in the special case of the full grid $\mathbb{F}_q^r$; see Bishnoi, Clark, Potukuchi and Schmitt [2] for the general “footprint” form. The writeup’s self-contained proof is kept: reduce P modulo $X_i^q - X_i$ to a polynomial R of the same function with all partial degrees $\leq q - 1$; by induction on r , expanding in the last variable and using that a univariate polynomial of degree a_r has at least $q - a_r$ non-zeros, $|\text{supp } R| \geq \prod_i (q - a_i)$ for some $0 \leq a_i \leq q - 1$ with $\sum a_i \leq \deg R \leq d$; finally $q - a \geq q^{1-a/(q-1)}$ by concavity of $\log$. [\[full proof\]](#)

Lemma 2.2 (Unit-valued lifts, folklore). *For every $\mathbb{F}_2$ -flow x on an oriented graph G there is an $\mathbb{F}_3$ -flow y with $\text{supp } y = \text{supp } x$.*

Idea of proof. The support of a binary flow is an even subgraph (every vertex has even degree in it, loops counting twice). Give each component an Eulerian circuit and set $y(e) = \pm 1$ according to whether the circuit traverses e along or against its orientation, $y = 0$ off the support. This is an integer circulation, hence an $\mathbb{F}_3$ -flow, with the required support. [\[full proof\]](#)

3 The group $\mathbb{Z}_7$

Proposition 3.1. *For every 3-edge-connected oriented graph G on $n \geq 2$ vertices and m edges and every $f: E \rightarrow \mathbb{Z}_7$,*

$$N_{\mathbb{Z}_7}(G, f) \geq 7^{r-m/6} = 7^{5m/6-n+1} \geq 7^{n/4+1} \geq (7^{1/4})^n.$$

Idea of proof. Parametrise the r -dimensional $\mathbb{F}_7$ -flow space by $\mathbb{F}_7^r$ and let L_e be the linear coordinate “value on e ”. The polynomial $P = \prod_e (L_e - f(e))$ has degree m and is non-zero at some point by [Theorem 1.2](#); [Lemma 2.1](#) with $q = 7$ gives $7^{r-m/6}$ non-zeros, each an f -avoiding flow. Then $r - m/6 = 5m/6 - n + 1 \geq n/4 + 1$ by (1). [\[full proof\]](#)

4 The group $\mathbb{Z}_6$

Fix a 3-edge-connected oriented G on $n \geq 2$ vertices, $f: E \rightarrow \mathbb{Z}_6$, and one f -avoiding flow ϕ_0 , which exists by [Theorem 1.2](#). Translation by ϕ_0 is a bijection between f -avoiding flows and flows ψ with

$$\psi(e) \neq t(e) \quad (e \in E), \quad t(e) := f(e) - \phi_0(e) \neq 0. \quad (2)$$

Identify $\mathbb{Z}_6 \cong \mathbb{F}_2 \times \mathbb{F}_3$ by the Chinese remainder theorem, so a $\mathbb{Z}_6$ -flow is a pair (x, y) of an $\mathbb{F}_2$ -flow and an $\mathbb{F}_3$ -flow, and write $t(e) = (t_2(e), t_3(e))$. The five non-zero elements of $\mathbb{Z}_6$ fall into three classes, and so do the edges:

$$\begin{aligned} A &= \{e : t(e) = (1, 0)\} \quad (\text{i.e. } t(e) = 3), \\ B &= \{e : t(e) = (0, \pm 1)\} \quad (\text{i.e. } t(e) \in \{2, 4\}), \\ C &= \{e : t(e) = (1, \pm 1)\} \quad (\text{i.e. } t(e) \in \{1, 5\}), \end{aligned} \quad (3)$$

with $a = |A|$, $b = |B|$, $c = |C|$, $a + b + c = m$.

Lemma 4.1 (Three families). *With the notation above, the number of flows ψ satisfying (2) is at least each of*

$$2^{(r-a)_+}, \quad 3^{(2r-b)_+/2}, \quad 2^{(r-c)_+}.$$

Idea of proof. Family 1: $\psi = (x, 0)$ with x a binary flow vanishing on A ; the only non-zero value is 3, which differs from t on $B \cup C$ (non-zero $\mathbb{F}_3$ -part) and is never taken on A . The binary flows vanishing on A form a subspace of dimension $\geq r - a$. Family 2: $\psi = (0, y)$ with y a ternary flow satisfying $y(e) \neq t_3(e)$ on B ; on $A \cup C$ the $\mathbb{F}_2$ -part already differs. Apply [Lemma 2.1](#) over $\mathbb{F}_3$ to $\prod_{e \in B} (L_e - t_3(e))$, degree b , non-zero at $y = 0$. Family 3: take a binary flow x vanishing on C and its lift y from [Lemma 2.2](#); then (x, y) takes only the unit values 1, 5 on its support, which avoid the non-units $t \in \{2, 3, 4\}$ on $A \cup B$, and is 0 on C where $t \neq 0$. Distinct x give distinct ψ . [\[full proof\]](#)

Proposition 4.2. *For every 3-edge-connected oriented graph G on $n \geq 2$ vertices and m edges and every $f: E \rightarrow \mathbb{Z}_6$,*

$$N_{\mathbb{Z}_6}(G, f) \geq 3^{(4r-m)/6} = 3^{(3m-4n+4)/6} \geq 3^{n/12+2/3} \geq (3^{1/12})^n.$$

Idea of proof. Put $d_A = (r - a)_+$, $d_B = (2r - b)_+$, $d_C = (r - c)_+$. Then $d_A + d_B + d_C \geq 4r - m$, so $\max\{d_A, d_B, d_C\} \geq (4r - m)/3$, and since $2^d \geq 3^{d/2}$ each family of [Lemma 4.1](#) has size at least $3^{d/2}$ for its own d . Substituting $r = m - n + 1$ and (1) gives the bound. [\[full proof\]](#)

5 Groups of order at least 8: a uniform constant

Repair (added in the rewrite). The writeup let $c_{\geq 8} > 1$ be “the uniform constant furnished by Theorem 1.9 of the source for groups of order at least 8”. As the referee observed, [Theorem 1.3](#) alone does not furnish one: for $n \leq 3$ its bound can be ≤ 1 (two vertices joined by three parallel edges have $\ell = 1$, and for $k = 8$ the bound is $\frac{1}{2} \cdot 2 = 1$). This section supplies the missing case analysis. The source paper’s own statement that Theorem 1.9 “shows the conjecture is true for all abelian groups except $\mathbb{Z}_6$ and $\mathbb{Z}_7$ ” has the same small- n slack, so the gap was inherited rather than introduced.

Lemma 5.1 (Parallel edges in small graphs). *A 3-edge-connected graph with $n \in \{2, 3\}$ vertices has two parallel (non-loop) edges e_1, e_2 joining the same pair of vertices.*

Idea of proof. For $n = 2$ every non-loop edge joins the two vertices and the single cut has at least three of them. For $n = 3$ the three singleton cuts have at least three edges each and every non-loop edge lies in exactly two of them, so there are at least five non-loop edges on three vertex pairs. [\[full proof\]](#)

Lemma 5.2 (Shifting along a 2-cycle). *Let G be an oriented graph with two parallel edges e_1, e_2 between u and v , let Γ be an abelian group, $f: E \rightarrow \Gamma$, and let ϕ_0 be an f -avoiding Γ -flow. Then $N_\Gamma(G, f) \geq |\Gamma| - 2$.*

Idea of proof. Let χ be the signed indicator of the closed walk $u \xrightarrow{e_1} v \xrightarrow{e_2} u$ ($\chi(e_i) = \pm 1$ according to orientation, 0 elsewhere); it is a flow. The flows $\phi_t = \phi_0 + t\chi$, $t \in \Gamma$, are pairwise distinct and agree with ϕ_0 off $\{e_1, e_2\}$; for each i , $\phi_t(e_i) = f(e_i)$ for exactly one t . So at least $|\Gamma| - 2$ of them are f -avoiding. [\[full proof\]](#)

Proposition 5.3. *Let G be a 3-edge-connected oriented graph on $n \geq 2$ vertices, Γ an abelian group with $|\Gamma| \geq 8$, and $f: E \rightarrow \Gamma$. Then*

$$N_\Gamma(G, f) \geq \begin{cases} \frac{1}{2}(3/2)^{n/2} & \text{if } n \geq 4, \\ |\Gamma| - 2 \geq 6 & \text{if } n \in \{2, 3\}, \end{cases} \quad \text{and in all cases} \quad N_\Gamma(G, f) \geq (2^{-1/4}(3/2)^{1/2})^n.$$

Idea of proof. For $k = |\Gamma| \geq 8$ the base in [Theorem 1.3](#) is $(k-4)/2 \geq 2$ (k even) or $(k-6)/2 \geq 3/2$ (k odd), so $N \geq \frac{1}{2}(3/2)^\ell$ with $\ell = m - n \geq n/2$ by (1). For $n \geq 4$, $\frac{1}{2}(3/2)^{n/2} \geq 2^{-n/4}(3/2)^{n/2}$ because $2^{-1} \geq 2^{-n/4}$. For $n \in \{2, 3\}$ use [Lemmas 5.1](#) and [5.2](#) with ϕ_0 from [Theorem 1.2](#): $N \geq |\Gamma| - 2 \geq 6 > 1.03^3$. [\[full proof\]](#)

6 Proof of the main theorem

Proof of Theorem 1.5. Parts (i), (ii) and (iii) are [Proposition 3.1](#), [Proposition 4.2](#) and [Proposition 5.3](#). For the final statement let $c = 2^{-1/4}(3/2)^{1/2}$; numerically $c \approx 0.8409 \cdot 1.2247 \approx 1.0299$, while $7^{1/4} \approx 1.6266$ and $3^{1/12} \approx 1.0959$, so $c = \min\{7^{1/4}, 3^{1/12}, c\}$. If $|\Gamma| = 7$ then $\Gamma \cong \mathbb{Z}_7$ and $N \geq (7^{1/4})^n \geq c^n$ by (i); if $|\Gamma| = 6$ then $\Gamma \cong \mathbb{Z}_6$ (the only abelian group of order 6) and $N \geq (3^{1/12})^n \geq c^n$ by (ii); if $|\Gamma| \geq 8$ then $N \geq c^n$ by (iii). Group connectivity depends on Γ only up to isomorphism, so this covers all abelian groups with $|\Gamma| \geq 6$. $\square$

7 Remarks

Remark 7.1 (Degenerate convention). The edgeless graph on one vertex is vacuously 3-edge-connected under a literal reading, has exactly one flow (the empty map), and would falsify [Conjecture 1.4](#) for every $c > 1$ ($1 < c^1$). The standard convention in this setting, used by the writeup and adopted here, is that 3-edge-connected graphs have at least two vertices; the source paper's Theorem 1.9 is likewise only meaningful under it (for the one-vertex graph its bound is $\frac{1}{2}$). Loops and parallel edges are allowed throughout, as in the source, and the proofs use nothing that excludes them: loops are free coordinates of the flow space and are handled by [Lemma 2.2](#) trivially.

Remark 7.2 (What is new). [Lemma 2.1](#) is the Alon–Füredi theorem and [Lemma 2.2](#) is folklore; [Theorems 1.2](#) and [1.3](#) are quoted. The new content is the application of [Lemma 2.1](#) to the flow polynomial for $\mathbb{Z}_7$ ([Proposition 3.1](#)) and the three-family argument for $\mathbb{Z}_6$ ([Proposition 4.2](#)). Both are short and elementary. The referee's report flags this as a reason for caution: the tools were available to the authors of [\[3\]](#), who instead obtained only a superpolynomial bound for $\mathbb{Z}_6$, and a six-year-old expert conjecture being settled by a five-line argument is a base-rate warning even when no error is found. Independent expert review is advised.

Remark 7.3 (Computational checks by the referee). The referee ([Appendix B](#)) verified [Lemma 2.1](#) exhaustively over all reduced polynomials for $(q, r) \in \{(2, 2), (2, 3), (3, 2)\}$ and on 350,000 random instances for $q \leq 7$, with the bound attained in tight cases; enumerated the complete $\mathbb{Z}_k$ -flow space on ten 3-edge-connected graphs (including multigraphs and a graph with a loop) and minimised N over all f on eight graph/group pairs, finding the bounds of [Propositions 3.1](#) and [4.2](#) satisfied with margin (e.g. for the triple edge over $\mathbb{Z}_7$, $\min_f N = 30 \geq 7^{3/2} \approx 18.5$); and implemented the three families of [Lemma 4.1](#) literally, asserting member-by-member admissibility and the claimed sizes on the same graphs. No violation was found.

Remark 7.4 (Novelty). The referee's literature search found only two papers citing [3], neither resolving [Conjecture 1.4](#), and no polynomial-method attack on group connectivity. This has not been checked by a human.

A Full proofs

Lemma 2.1 (Alon–Füredi support bound). *Let q be a prime power and let $P \in \mathbb{F}_q[X_1, \dots, X_r]$ induce a non-zero function on $\mathbb{F}_q^r$, with $\deg P \leq d$. Then*

$$|\{x \in \mathbb{F}_q^r : P(x) \neq 0\}| \geq q^{(r-d/(q-1))_+}, \quad \text{where } (z)_+ = \max\{z, 0\}.$$

Proof. Since $x^q = x$ for all $x \in \mathbb{F}_q$, reducing P modulo the polynomials $X_i^q - X_i$ yields $R \in \mathbb{F}_q[X_1, \dots, X_r]$ inducing the same function as P , with $\deg_{X_i} R \leq q - 1$ for each i and $\deg R \leq \deg P \leq d$ (reduction never increases total degree). As the function is non-zero, $R \neq 0$.

Claim. There are integers $0 \leq a_i \leq q - 1$ with $\sum_i a_i \leq \deg R$ such that $|\text{supp } R| \geq \prod_{i=1}^r (q - a_i)$. We argue by induction on r . For $r = 0$, R is a non-zero constant and the claim is trivial. For $r \geq 1$ write $R = \sum_{j=0}^{a_r} R_j(X_1, \dots, X_{r-1})X_r^j$ with $R_{a_r} \neq 0$ and $a_r \leq q - 1$; then $\deg R_{a_r} \leq \deg R - a_r$. For every $x' \in \mathbb{F}_q^{r-1}$ with $R_{a_r}(x') \neq 0$, the univariate polynomial $R(x', X_r)$ has degree exactly a_r , hence at most a_r roots, hence at least $q - a_r$ non-zeros. By induction applied to R_{a_r} there are $a_1, \dots, a_{r-1}$ with $\sum_{i < r} a_i \leq \deg R_{a_r} \leq \deg R - a_r$ and $|\text{supp } R_{a_r}| \geq \prod_{i < r} (q - a_i)$. Hence $|\text{supp } R| \geq (q - a_r) \prod_{i < r} (q - a_i)$ with $\sum_{i \leq r} a_i \leq \deg R$.

For $0 \leq a \leq q - 1$ the function $a \mapsto \log(q - a)$ is concave on $[0, q - 1]$, takes the value $\log q$ at $a = 0$ and 0 at $a = q - 1$, hence lies above the chord: $\log(q - a) \geq (1 - a/(q - 1)) \log q$, i.e. $q - a \geq q^{1-a/(q-1)}$. Therefore

$$|\text{supp } R| \geq \prod_i q^{1-a_i/(q-1)} = q^{r - \sum_i a_i/(q-1)} \geq q^{r-d/(q-1)}.$$

If $r - d/(q - 1) < 0$ the claimed bound is $q^0 = 1$, which holds because $R \neq 0$ as a function. $\square$

Lemma 2.2 (Unit-valued lifts, folklore). *For every $\mathbb{F}_2$ -flow x on an oriented graph G there is an $\mathbb{F}_3$ -flow y with $\text{supp } y = \text{supp } x$.*

Proof. Let $S = \text{supp } x$. Kirchhoff's law over $\mathbb{F}_2$ at a vertex v says that the number of non-loop edges of S incident with v is even (a loop contributes $x(e)$ to both sides and cancels). Thus in the subgraph (V, S) every vertex has even degree, loops counting twice, so each component of (V, S) has an Eulerian circuit. Traverse each such circuit and define $y(e) = +1$ if the circuit traverses e in the direction of its orientation in G , $y(e) = -1$ if against, and $y(e) = 0$ for $e \notin S$ (a loop is traversed once, in one of the two directions). Along a closed walk every vertex is entered as often as it is left, so y is an integer circulation: at every vertex the sum of y over outgoing edges equals the sum over incoming edges. Reducing modulo 3 gives an $\mathbb{F}_3$ -flow, and $\text{supp } y = S$ since $\pm 1 \neq 0$ in $\mathbb{F}_3$. $\square$

Proposition 3.1. *For every 3-edge-connected oriented graph G on $n \geq 2$ vertices and m edges and every $f: E \rightarrow \mathbb{Z}_7$,*

$$N_{\mathbb{Z}_7}(G, f) \geq 7^{r-m/6} = 7^{5m/6-n+1} \geq 7^{n/4+1} \geq (7^{1/4})^n.$$

Proof. G is connected (it is 3-edge-connected with $n \geq 2$), so the $\mathbb{F}_7$ -flows on G form a vector space of dimension $r = m - n + 1$ (the cycle space; loops are free coordinates and are counted in m). Fix a linear isomorphism $\mathbb{F}_7^r \rightarrow \{\mathbb{F}_7\text{-flows}\}$, $X \mapsto \phi_X$, and for each edge e let $L_e(X) = \phi_X(e)$, a linear form in X . Put

$$P(X) = \prod_{e \in E} (L_e(X) - f(e)) \in \mathbb{F}_7[X_1, \dots, X_r], \quad \deg P \leq m.$$

Then $P(X) \neq 0$ if and only if $\phi_X(e) \neq f(e)$ for every e , i.e. iff ϕ_X is f -avoiding, and $X \mapsto \phi_X$ is a bijection, so $N_{\mathbb{Z}_7}(G, f) = |\{X : P(X) \neq 0\}|$. By [Theorem 1.2](#) some f -avoiding flow exists, so P is non-zero as a function. [Lemma 2.1](#) with $q = 7$ and $d = m$ gives

$$N_{\mathbb{Z}_7}(G, f) \geq 7^{r-m/6} = 7^{m-n+1-m/6} = 7^{5m/6-n+1}.$$

The exponent is increasing in m , and $m \geq \frac{3}{2}n$ by (1), so $5m/6 - n + 1 \geq \frac{5}{4}n - n + 1 = n/4 + 1 > 0$ (in particular the positive part in [Lemma 2.1](#) is not active). Finally $7^{n/4+1} \geq 7^{n/4} = (7^{1/4})^n$. $\square$

Lemma 4.1 (Three families). *With the notation above, the number of flows ψ satisfying (2) is at least each of*

$$2^{(r-a)_+}, \quad 3^{(2r-b)_+/2}, \quad 2^{(r-c)_+}.$$

Proof. Throughout, a pair (x, y) with x an $\mathbb{F}_2$ -flow and y an $\mathbb{F}_3$ -flow is a $\mathbb{Z}_6$ -flow under $\mathbb{Z}_6 \cong \mathbb{F}_2 \times \mathbb{F}_3$, and it satisfies (2) iff $(x(e), y(e)) \neq (t_2(e), t_3(e))$ for every e . Recall from (3): on A , $t = (1, 0)$; on B , $t = (0, \pm 1)$; on C , $t = (1, \pm 1)$.

Family 1. Let x be an $\mathbb{F}_2$ -flow with $x(e) = 0$ for all $e \in A$, and $\psi = (x, 0)$. On $e \in A$: $\psi(e) = (0, 0) \neq (1, 0) = t(e)$. On $e \in B \cup C$: the second coordinate of $\psi(e)$ is 0 while $t_3(e) = \pm 1 \neq 0$. So ψ satisfies (2). The $\mathbb{F}_2$ -flows form an r -dimensional space and the a linear conditions $x(e) = 0$ ($e \in A$) have rank at most a , so there are at least $2^{\max\{r-a, 0\}} = 2^{(r-a)_+}$ such x , giving as many distinct ψ .

Family 2. Let y be an $\mathbb{F}_3$ -flow with $y(e) \neq t_3(e)$ for all $e \in B$, and $\psi = (0, y)$. On $A \cup C$: the first coordinate of $\psi(e)$ is 0 while $t_2(e) = 1$. On B : $\psi(e) = (0, y(e)) \neq (0, t_3(e))$ by choice. So ψ satisfies (2). Parametrise the r -dimensional $\mathbb{F}_3$ -flow space by $\mathbb{F}_3^r$ with coordinate forms L_e and consider $Q = \prod_{e \in B} (L_e - t_3(e))$, of degree b ; $Q(0) = \prod_{e \in B} (-t_3(e)) \neq 0$ since $t_3 \neq 0$ on B . By [Lemma 2.1](#) with $q = 3$, $d = b$, the number of admissible y is at least $3^{(r-b/2)_+} = 3^{(2r-b)_+/2}$.

Family 3. Let x be an $\mathbb{F}_2$ -flow with $x(e) = 0$ for all $e \in C$, and let y be the $\mathbb{F}_3$ -flow of [Lemma 2.2](#) with $\text{supp } y = \text{supp } x$; put $\psi = (x, y)$. For $e \in \text{supp } x$, $\psi(e) = (1, \pm 1)$, i.e. $\psi(e) \in \{1, 5\}$ is a unit of $\mathbb{Z}_6$; for $e \notin \text{supp } x$, $\psi(e) = 0$. On $e \in C$: $x(e) = 0$, so $\psi(e) = 0 \neq t(e)$. On $e \in A \cup B$: $t(e) \in \{2, 3, 4\}$ is a non-unit and non-zero, so neither a unit nor 0 equals it. So ψ satisfies (2). Distinct x give distinct ψ (they differ in the first coordinate), and as in Family 1 there are at least $2^{(r-c)_+}$ admissible x . $\square$

Proposition 4.2. *For every 3-edge-connected oriented graph G on $n \geq 2$ vertices and m edges and every $f: E \rightarrow \mathbb{Z}_6$,*

$$N_{\mathbb{Z}_6}(G, f) \geq 3^{(4r-m)/6} = 3^{(3m-4n+4)/6} \geq 3^{n/12+2/3} \geq (3^{1/12})^n.$$

Proof. Translation by ϕ_0 shows $N_{\mathbb{Z}_6}(G, f)$ equals the number of flows satisfying (2), so by [Lemma 4.1](#)

$$N_{\mathbb{Z}_6}(G, f) \geq \max\{2^{d_A}, 3^{d_B/2}, 2^{d_C}\}, \quad d_A = (r-a)_+, \quad d_B = (2r-b)_+, \quad d_C = (r-c)_+.$$

Since $(z)_+ \geq z$ and $a + b + c = m$,

$$d_A + d_B + d_C \geq (r-a) + (2r-b) + (r-c) = 4r - m,$$

so $d^* := \max\{d_A, d_B, d_C\} \geq (4r - m)/3$. As $2^d \geq 3^{d/2}$ for $d \geq 0$ (because $4 \geq 3$), each of the three lower bounds is at least $3^{d^*/2}$ for its own d , hence $N_{\mathbb{Z}_6}(G, f) \geq 3^{d^*/2} \geq 3^{(4r-m)/6}$. Substituting $r = m - n + 1$ gives $4r - m = 3m - 4n + 4$, and by (1) $3m - 4n + 4 \geq \frac{9}{2}n - 4n + 4 = \frac{n}{2} + 4 > 0$. Therefore

$$N_{\mathbb{Z}_6}(G, f) \geq 3^{(3m-4n+4)/6} \geq 3^{n/12+2/3} \geq (3^{1/12})^n. \quad \square$$

Lemma 5.1 (Parallel edges in small graphs). *A 3-edge-connected graph with $n \in \{2, 3\}$ vertices has two parallel (non-loop) edges e_1, e_2 joining the same pair of vertices.*

Proof. Loops lie in no edge cut, so only non-loop edges matter for edge-connectivity. If $n = 2$, every non-loop edge joins the two vertices u, v , and the cut $\{u\} \mid \{v\}$ contains at least three of them; any two are parallel. If $n = 3$ with vertices u, v, w , each of the cuts $\{u\} \mid \{v, w\}$, $\{v\} \mid \{u, w\}$, $\{w\} \mid \{u, v\}$ contains at least three non-loop edges, and each non-loop edge lies in exactly two of these three cuts; hence $2m' \geq 9$ for the number m' of non-loop edges, so $m' \geq 5$. Five edges on the three pairs $\{u, v\}$, $\{u, w\}$, $\{v, w\}$ include two on the same pair. $\square$

Lemma 5.2 (Shifting along a 2-cycle). *Let G be an oriented graph with two parallel edges e_1, e_2 between u and v , let Γ be an abelian group, $f: E \rightarrow \Gamma$, and let ϕ_0 be an f -avoiding Γ -flow. Then $N_\Gamma(G, f) \geq |\Gamma| - 2$.*

Proof. Define $\chi: E \rightarrow \mathbb{Z}$ by $\chi(e_1) = +1$ if e_1 is oriented from u to v and -1 otherwise, $\chi(e_2) = -1$ if e_2 is oriented from u to v and $+1$ otherwise, and $\chi(e) = 0$ for all other edges. Then χ is a circulation: at u the net outflow contributed by e_1 is $+1$ and by e_2 is -1 (in both orientation cases), and symmetrically at v ; all other vertices are untouched. For $t \in \Gamma$ put $\phi_t = \phi_0 + t\chi$ (i.e. $\phi_t(e) = \phi_0(e) + \chi(e)t$). Each ϕ_t is a Γ -flow, being the sum of a flow and t times a circulation. The map $t \mapsto \phi_t(e_1) = \phi_0(e_1) \pm t$ is a bijection $\Gamma \rightarrow \Gamma$, so the ϕ_t are pairwise distinct, and $\phi_t(e_1) = f(e_1)$ for exactly one t ; likewise $\phi_t(e_2) = f(e_2)$ for exactly one t . For $e \notin \{e_1, e_2\}$, $\phi_t(e) = \phi_0(e) \neq f(e)$. Hence at least $|\Gamma| - 2$ of the flows ϕ_t are f -avoiding. $\square$

Proposition 5.3. *Let G be a 3-edge-connected oriented graph on $n \geq 2$ vertices, Γ an abelian group with $|\Gamma| \geq 8$, and $f: E \rightarrow \Gamma$. Then*

$$N_\Gamma(G, f) \geq \begin{cases} \frac{1}{2}(3/2)^{n/2} & \text{if } n \geq 4, \\ |\Gamma| - 2 \geq 6 & \text{if } n \in \{2, 3\}, \end{cases} \quad \text{and in all cases} \quad N_\Gamma(G, f) \geq (2^{-1/4}(3/2)^{1/2})^n.$$

Proof. Let $k = |\Gamma| \geq 8$ and $\ell = m - n$, so $\ell \geq n/2$ by (1), and $\ell \geq 1$ since $n \geq 2$.

Case $n \geq 4$. If k is even then $k \geq 8$ and $(k - 4)/2 \geq 2 \geq 3/2$; if k is odd then $k \geq 9$ and $(k - 6)/2 \geq 3/2$. In both cases Theorem 1.3 gives $N_\Gamma(G, f) \geq \frac{1}{2}(3/2)^\ell \geq \frac{1}{2}(3/2)^{n/2}$. For $n \geq 4$ we have $2^{-1} \geq 2^{-n/4}$, hence

$$\frac{1}{2}(3/2)^{n/2} \geq 2^{-n/4}(3/2)^{n/2} = (2^{-1/4}(3/2)^{1/2})^n.$$

(For $n \leq 3$ the bound $\frac{1}{2}(3/2)^{n/2}$ is at most $\frac{1}{2}(3/2)^{3/2} < 1$, which is why this case needs a separate argument.)

Case $n \in \{2, 3\}$. By Theorem 1.2 there is an f -avoiding flow ϕ_0 , and by Lemma 5.1 G has two parallel edges. Lemma 5.2 gives $N_\Gamma(G, f) \geq |\Gamma| - 2 \geq 6$, and $6 > (1.03)^3 > (2^{-1/4}(3/2)^{1/2})^n$ for $n \leq 3$ since $2^{-1/4}(3/2)^{1/2} < 1.03$. $\square$

Theorem 1.5 (Main theorem). *Let $G = (V, E)$ be a 3-edge-connected oriented graph with $n \geq 2$ vertices and m edges, and let $f: E \rightarrow \Gamma$.*

- (i) *If $\Gamma = \mathbb{Z}_7$ then $N_{\mathbb{Z}_7}(G, f) \geq 7^{5m/6-n+1} \geq 7^{n/4+1}$.*
- (ii) *If $\Gamma = \mathbb{Z}_6$ then $N_{\mathbb{Z}_6}(G, f) \geq 3^{(3m-4n+4)/6} \geq 3^{n/12+2/3}$.*
- (iii) *If $|\Gamma| \geq 8$ then $N_\Gamma(G, f) \geq \frac{1}{2}(3/2)^{n/2}$ when $n \geq 4$, and $N_\Gamma(G, f) \geq |\Gamma| - 2 \geq 6$ when $n \in \{2, 3\}$.*

Consequently $N_\Gamma(G, f) \geq c^n$ for every abelian group Γ with $|\Gamma| \geq 6$, with the absolute constant

$$c = \min\{7^{1/4}, 3^{1/12}, 2^{-1/4}(3/2)^{1/2}\} = 2^{-1/4}(3/2)^{1/2} \approx 1.0299.$$

Proof. See Section 6; parts (i)–(iii) are Propositions 3.1, 4.2 and 5.3, proved above. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the *original* writeup. Its step numbers refer to that writeup, not to this note; its step 12 is the gap repaired in [Section 5](#). The scripts it mentions are in `verification/scripts/2005.09767__00/`. Treat it as machine output, not as an authority.

Target claim: **proved**. The writeup claims exponentially many (c^n) Γ -flows avoiding an arbitrary forbidden function f on every 3-edge-connected oriented graph, for every abelian Γ with $|\Gamma| \geq 6$, by settling the two cases left open by DeVos–Langhede–Mohar–Šámal (arXiv:2005.09767): $\mathbb{Z}_7$ and $\mathbb{Z}_6$.

B.1 Interpretation

The writeup attacks exactly the catalog statement, which I confirmed verbatim against the arXiv source (Conjecture 1.10 of arXiv:2005.09767): fixed $c > 1$, all 3-edge-connected oriented graphs of order n , all abelian Γ with $|\Gamma| \geq 6$, all $f : E \rightarrow \Gamma$, at least c^n flows ϕ with $\phi(e) \neq f(e)$ everywhere. There is no interpretation gaming:

- n is the number of vertices (order), used correctly throughout; the final bounds are $7^{n/4+1}$ for $\mathbb{Z}_7$ and $3^{n/12+2/3}$ for $\mathbb{Z}_6$, both of the required c^n shape.
- The setting is multigraphs (as required in flow theory); the arguments nowhere assume simplicity, and my computational checks include parallel edges and a loop.
- The reduction to $\mathbb{Z}_6, \mathbb{Z}_7$ via the classification of abelian groups of order 6 and 7 is correct ($\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$; order 7 is prime), and group connectivity depends on Γ only up to isomorphism.
- The degenerate one-vertex edgeless graph is explicitly excluded by the standard convention, which the writeup itself flags (its caveat 3). A reasonable author of the source paper would consider this the intended conjecture.

B.2 Step-by-step findings

step	label	note
1. Setup and reduction to $\mathbb{Z}_6, \mathbb{Z}_7$	VALID	Flow-space/cycle-rank conventions correct ($r = m - n + 1$ for connected multigraphs, over any field, loops included). Reduction uses Theorem 1.9 (confirmed in source) for $ \Gamma \geq 8$ and group classification for orders 6, 7. The uniformity of the constant for $ \Gamma \geq 8$ has a small- n wrinkle deferred to step 12.

step	label	note
2. Lemma 1 (support/footprint bound $\geq q^{(r-d/(q-1))_+}$)	VALID	This is exactly the known Alon–Füredi grid bound (Alon–Füredi 1993; cf. Bishnoi–Clark–Potukuchi–Schmitt, arXiv:1508.06020), stated without attribution but proved. I re-derived the induction (support $\geq \prod (q - a_i)$ with $\sum a_i \leq \deg R$) and the concavity step $q - a \geq q^{1-a/(q-1)}$ (equality at $a = 0$ and $a = q - 1$, concavity of $\log(q - a)$) in full; both are correct. Verified computationally: exhaustive over all reduced polynomials for $(q, r) \in \{(2, 2), (2, 3), (3, 2)\}$ and 350k randomized cases for $q \leq 7$, zero violations, with tight cases attaining ratio exactly 1.
3. $\mathbb{Z}_7$: parametrization of the flow space	VALID	$\mathbb{F}_7$ -flow space has dimension $r = m - n + 1$; coordinate functions L_e are linear forms. Standard.
4. $\mathbb{Z}_7$: $P = \prod_e (L_e - f(e))$, $\deg P \leq m$, $P \not\equiv 0$ as a function	VALID	Nonvanishing at one point is exactly the Jaeger–Linial–Payan–Tarsi theorem (3-edge-connected $\Rightarrow \Gamma$ -connected for $ \Gamma \geq 6$), confirmed as quoted in the source paper.
5. $\mathbb{Z}_7$: $N \geq 7^{r-m/6} = 7^{5m/6-n+1}$	VALID	Direct application of Lemma 1 with $q = 7$, $d = m$; exponent positive since $m \geq 3n/2$. Re-derived; consistent with exhaustive small-graph minima (see computational check).
6. $\mathbb{Z}_7$: $m \geq 3n/2$ and $N \geq 7^{n/4+1} \geq (7^{1/4})^n$	VALID	$2m = \sum \deg \geq 3n$ for 3-edge-connected multigraphs; exponent $5m/6 - n + 1$ is increasing in m , minimized at $m = 3n/2$ giving $n/4 + 1$. Arithmetic checked.

step	label	note
7. $\mathbb{Z}_6$: normalization $t = f - \phi_0$ (t nowhere zero), CRT split, classes A, B, C	VALID	ϕ_0 exists by JLPT; translation is a bijection of the flow group; CRT class bookkeeping ($3 \leftrightarrow (1, 0)$; $2, 4 \leftrightarrow (0, \pm 1)$; $1, 5 \leftrightarrow (1, \pm 1)$) checked.
8. Family 1 (flows $(x, 0)$, $x = 0$ on A): $\geq 2^{(r-a)_+}$ admissible	VALID	Avoidance argument on A , $B \cup C$ correct; dimension count correct. Machine-verified member-by-member on 10 graphs $\times$ 60 random f .
9. Family 2 (flows $(0, y)$, $y \neq t_3$ on B): $\geq 3^{(2r-b)_+/2}$ admissible	VALID	Lemma 1 over $\mathbb{F}_3$ with $d = b$, nonzero function witnessed by $y = 0$ (since $t_3 \neq 0$ on B). Machine-verified.
10. Lemma 2 (unit-valued $\mathbb{F}_3$ -lift of a binary flow) and Family 3: $\geq 2^{(r-c)_+}$ admissible	VALID	Support of a binary flow is an even subgraph; Eulerian orientation gives a ± 1 integer circulation with equal support (holds with loops and parallel edges). Units 1, 5 can never equal targets in $A \cup B$ (nonunits) and the support avoids C ; injectivity via the first CRT coordinate. Machine-verified, including an explicit Hierholzer implementation of Lemma 2.
11. Combination: $N \geq 3^{(4r-m)/6} =$ $3^{(3m-4n+4)/6} \geq 3^{n/12+2/3}$	VALID	$(z)_+ \geq z$, $a + b + c = m$, $\max \geq \text{average}$, $2^d \geq 3^{d/2}$, $r = m - n + 1$, $m \geq 3n/2$: all re-derived; arithmetic exact.

step	label	note
12. Completion: $c = \min\{c_{\geq 8}, 3^{1/12}, 7^{1/4}\}$	GAP (minor, repairable)	Theorem 1.9's actual bound is $\frac{1}{2}(\frac{k-6}{2})^\ell$ (k odd), $\frac{1}{2}(\frac{k-4}{2})^\ell$ (k even), with $\ell = m - n$. For $n \leq 3$ this can be ≤ 1 (e.g. the 2-vertex triple edge, $\ell = 1$, $k = 8$: bound = $1 < c^2$), so Theorem 1.9 as stated does not by itself “furnish” a constant $c_{\geq 8}$ valid for all n. The repair is routine: any 3-edge-connected graph on ≤ 3 vertices has a parallel pair, and shifting an avoiding flow along that 2-edge cycle yields $\geq \Gamma - 2 \geq 6$ avoiding flows, more than c^n for $n \leq 3$ and c small. Note this reading of Theorem 1.9 (“confirms the conjecture for $\Gamma \geq 8$”) is the source paper's own, so the slack is inherited, not introduced.

No step is an ERROR. The genuinely new content (steps 2–11) is fully valid as far as I can determine.

B.3 Reference check

1. **Conjecture 1.10, Theorem 1.9, Theorem 1.11** of arXiv:2005.09767 (DeVos, Langhede, Mohar, Šámal, “Many flows in the group connectivity setting”, v1, 19 May 2020, single version): fetched via arXiv/ar5iv; statements match the writeup's use exactly. Theorem 1.9: $N \geq \frac{1}{2}(\frac{k-6}{2})^\ell$ (k odd), $\frac{1}{2}(\frac{k-4}{2})^\ell$ (k even), $\ell = |E| - |V|$ — degenerate precisely for $k = 6, 7$, as the writeup says.
2. **Jaeger–Linial–Payan–Tarsi**: the source's abstract states “Jaeger et al. proved that every oriented 3-edge-connected graph is Γ -connected whenever $|\Gamma| \geq 6$ ”, and independent sources confirm JLPT (JCTB 1992) derive A -connectivity of all 3-edge-connected graphs for $|A| \geq 6$ from a cubic-graph decomposition. This backs both existence inputs (the $\mathbb{Z}_7$ nonvanishing point and the $\mathbb{Z}_6$ flow ϕ_0). Confirmed.
3. **Lemma 1** is presented as new but is the classical Alon–Füredi / footprint bound (Alon–Füredi, *Covering the cube by affine hyperplanes*, 1993; Bishnoi–Clark–Potukuchi–Schmitt, arXiv:1508.06020). Its proof in the writeup is self-contained and correct; missing attribution is a presentation issue, not a soundness issue — if anything it raises confidence, since the lemma is an established theorem.
4. **Prior resolution search**: Semantic Scholar lists only two papers citing 2005.09767 (Husek–Šámal, “Exponentially Many Circuit Double Covers”; Langhede–Thomassen, “Exponentially many $\mathbb{Z}_5$ -colorings in simple planar graphs”) — neither resolves Conjecture 1.10. Targeted searches for a polynomial-method/Alon–Füredi attack on group connectivity or a $\mathbb{Z}_6/\mathbb{Z}_7$ resolution (through September 2026) found nothing. I could not find this argument in the literature, so ALREADY_KNOWN does not apply as far as I can tell.

B.4 Computational check

Scripts under `verification/scripts/2005.09767__00/`:

- `check_lemma1.py` — Lemma 1: exhaustive over all reduced polynomials for $(q, r) = (2, 2), (2, 3), (3, 2)$ (15 + 255 + 19682 polynomials) and 350,000 randomized cases for $q \in \{3, 5, 7\}$, $r \leq 3$, plus 2000 random products of linear factors (the exact shape used in the proof). **0 violations**; minimum support/bound ratio 1.0000 (the bound is attained, so my encoding is not vacuously loose).
- `check_flows.py` — enumerates the full $\mathbb{Z}_k$ -flow space (k^r flows via a fundamental-cycle basis, conservation asserted) for ten 3-edge-connected graphs (2-vertex triple/quadruple edges, tripled triangle, K_4 , K_4 +loop, K_4 with a doubled edge, prism, $K_{3,3}$, wheel W_5 , Petersen) and minimizes $N(G, f)$ over **all** f where feasible ($k^m \leq 4 \cdot 10^5$: 8 graph/group pairs exhaustive) else over ~ 590 random+adversarial f . Claimed bounds hold everywhere, with margin. Sample exact numbers: triple edge $\mathbb{Z}_7$: $\min N = 30 \geq 7^{1.5} \approx 18.52$ (exhaustive over all 343 f); $K_4 \mathbb{Z}_7$: $\min N = 120 \geq 49$ (exhaustive over 7^6 f); triple edge $\mathbb{Z}_6$: $\min N = 20 \geq 3^{5/6} \approx 2.50$ (exhaustive); $K_4 \mathbb{Z}_6$: $\min N = 60 \geq 3$ (exhaustive over 6^6 f); Petersen: $\min$ sampled $N = 8640$ vs bound 907.5 ($\mathbb{Z}_7$), 1920 vs 5.20 ($\mathbb{Z}_6$). **0 violations**.
- `check_z6_families.py` — implements Section 3 literally: finds ϕ_0 by brute force, forms t and the classes A, B, C , constructs all three families (including an actual Eulerian-orientation lift for Lemma 2), asserts every constructed member is a valid $\mathbb{Z}_6$ -flow avoiding t , that the families meet their counts $2^{(r-a)_+}, 3^{(2r-b)_+/2}, 2^{(r-c)_+}$, and that $N \geq 3^{(4r-m)/6}$ — over the same ten graphs $\times$ 60 random f each. **All assertions pass**, including on the graphs with loops and parallel edges.

No claimed property failed computationally.

B.5 Caveats

1. **Small- n uniformity of $c_{\geq 8}$** (step 12): Theorem 1.9’s stated bound is ≤ 1 for some 3-edge-connected graphs with $n \leq 3$ (e.g. $n = 2, \ell = 1, k = 8$), so the “uniform constant furnished by Theorem 1.9” needs a two-line patch for $n \leq 3$ (shift along a parallel pair: $\geq |\Gamma| - 2$ flows). Routine; also inherent in the source paper’s own claim that Theorem 1.9 confirms the conjecture for $|\Gamma| \geq 8$. The writeup’s *new* bounds ($7^{n/4+1}, 3^{n/12+2/3}$) have no such slack: they are > 1 and of the form c^n for every $n \geq 2$.
2. **Degenerate conventions**: the one-vertex edgeless graph would falsify the literal statement ($N = 1 < c^1$); the writeup flags this itself, and the standard convention (3-edge-connected $\Rightarrow n \geq 2$) excludes it. With $n \geq 2$ everything goes through, loops and parallel edges included (checked).
3. **Attribution**: Lemma 1 is the Alon–Füredi/footprint bound and should be cited, not presented as new; Lemma 2 (Eulerian ± 1 lift) is folklore. Cosmetic.
4. **Extraordinariness**: if correct, this elementary argument (JLPT + Alon–Füredi for $\mathbb{Z}_7$; a short three-family CRT argument beating Theorem 1.11’s $2^{\sqrt{\ell}/\log \ell}$ by an exponential margin for $\mathbb{Z}_6$) resolves a conjecture that four experts left open in 2020 and that remained open through the catalog’s May 2026 review. The tools were fully available to the authors, which is a sociological red flag even though I found no mathematical error. Computational verification necessarily covers only small graphs; the general-case steps rest on my hand-verification. Independent expert review is strongly advised before announcing.
5. **Novelty is unverified**: I found no prior appearance of this argument or any resolution of Conjecture 1.10, but web/citation coverage of very recent preprints is imperfect.

B.6 Referee summary

I set out assuming the writeup was wrong and failed to break it. The interpretation is the intended one. The $\mathbb{Z}_7$ case is a five-line application of the (correct, classical, machine-re-verified) Alon–Füredi

footprint bound to the product polynomial $\prod_e (L_e - f(e))$ on the r -dimensional $\mathbb{F}_7$ -flow space, seeded by the JLPT existence theorem: $N \geq 7^{r-m/6} \geq 7^{n/4+1}$. The $\mathbb{Z}_6$ case translates by one admissible flow, splits $\mathbb{Z}_6 \cong \mathbb{F}_2 \times \mathbb{F}_3$, and produces three families of admissible flows whose sizes cannot all be small, giving $N \geq 3^{(3m-4n+4)/6} \geq 3^{n/12+2/3}$; I re-derived every inequality and verified the construction member-by-member in code on ten graphs (including multigraphs and a loop), with exhaustive minimization over all f on eight graph/group pairs — zero violations, and the bounds held with large margins. Both cited inputs (JLPT; Theorem 1.9 of the source) check out against the arXiv text. The single defect found is the small- n slack in deriving a uniform constant from Theorem 1.9 for $|\Gamma| \geq 8$ (step 12), which is trivially repairable and inherited from the source's own reading. Hence MINOR_GAPS rather than CONFIRMED; confidence is medium only because the claim, if correct, resolves a six-year-old expert conjecture with tools its authors possessed — a base-rate warning that no local check can fully discharge.

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